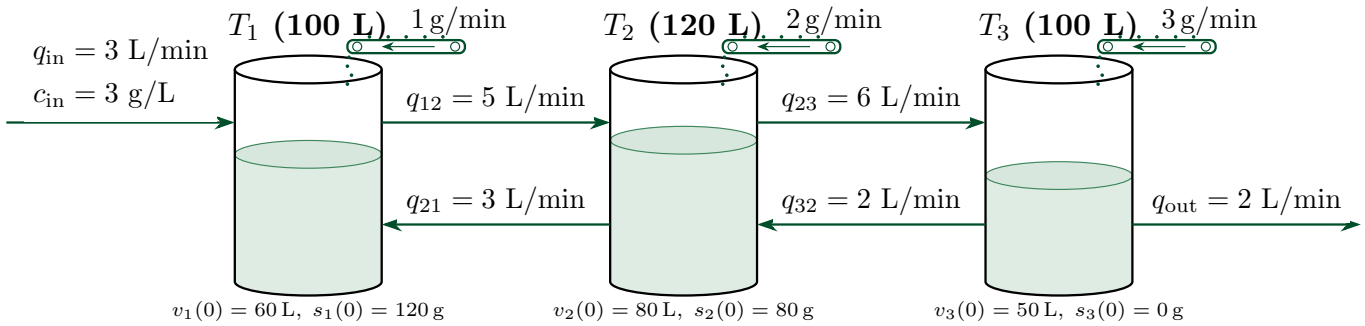
MA3521 — In-Class Tank Worksheet

Name:

1.



The tanks are well-mixed. Time t is in minutes, volumes $\mathbf{v}(t)$ in L, and salt masses $\mathbf{s}(t)$ in g.
The salt sources above the tanks add salt continuously without adding liquid. Include these rates in $\mathbf{b}$.
The IVP is $\mathbf{s}' = \boldsymbol{A}(t)\mathbf{s} + \mathbf{b}$, $\mathbf{s}(0) = \mathbf{s}_0$.

$0 < t <$ ____

$\mathbf{v}(t) =$ ____ L

$\mathbf{s}_0 =$ ____ g

$\mathbf{b} =$ ____ g/min

$\boldsymbol{A}(t) =$ ____ min^{-1}